

Class Diagrams

I made two class diagrams for this project. The program keeps the code that talks to the database separate from the data itself, so it made sense to draw each one on its own: the first shows the classes I actually wrote, the second shows the data they work with. Between them they cover everything in the scope — currencies, customers, rates, transactions and the two reports.

Diagram 1 — Application classes

This one shows how the program is put together. Database looks after the connection and creates the tables. Then there are four manager classes, one per table — currencies, customers, rates and transactions — and each of them is handed the same database to work with. The main module keeps one of each and just passes the menu choice along, which is why there's no SQL anywhere near the menu. The one extra link worth pointing out: TransactionManager also keeps an ExchangeRateManager, since it needs to look up the current rate before it can save an exchange.

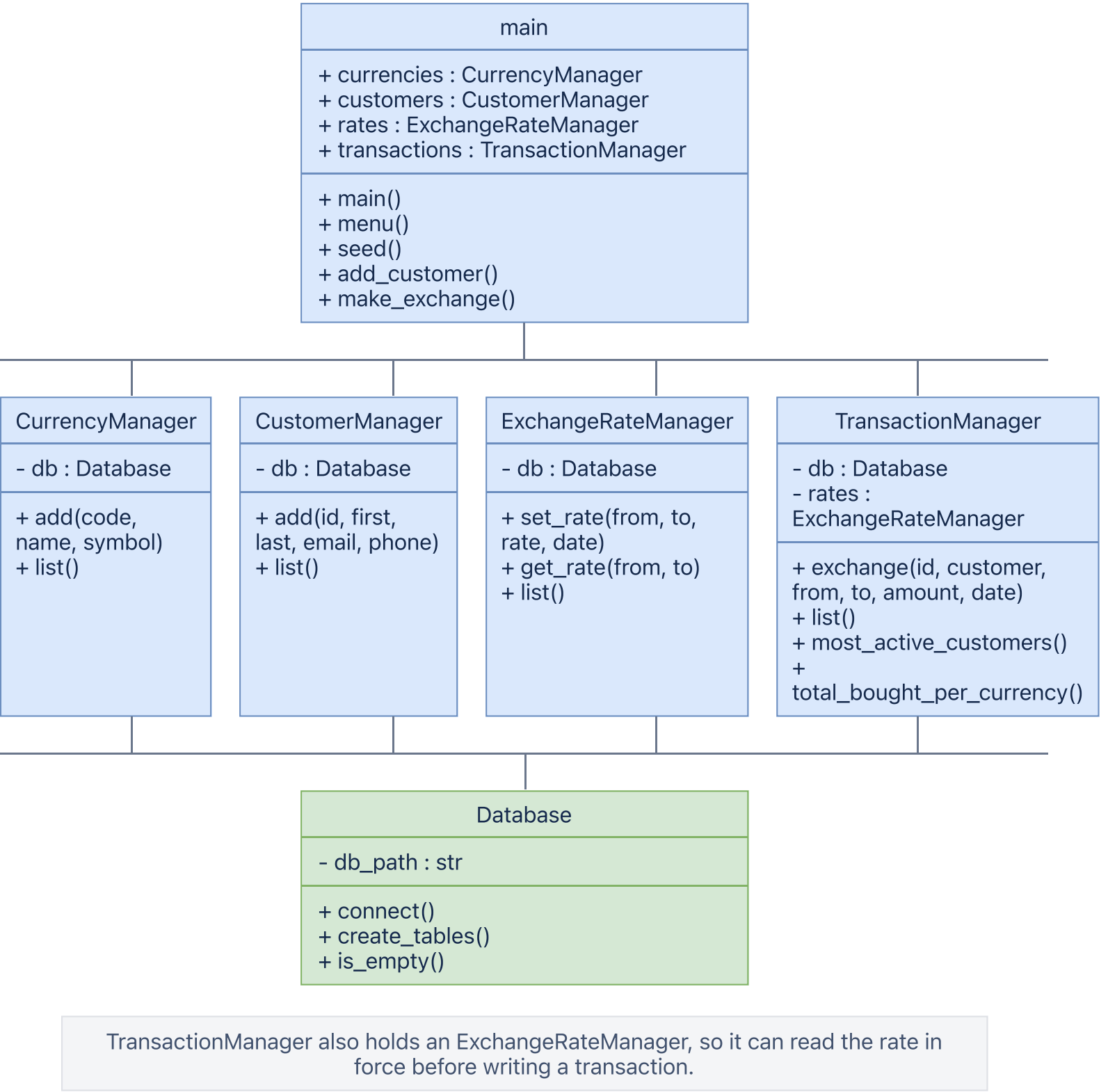


Diagram 2 — Domain classes

This one is the data itself, drawn as classes instead of tables. Currency sits in the middle because everything points back to it — a rate connects two currencies without really belonging to either, and a transaction names two currencies plus the customer it was made for. A customer can have plenty of transactions, but each transaction only ever has one customer. Note that the rate is copied onto the transaction rather than looked up again later, so an old exchange still shows the rate it was actually made at once the rates move on.

